

PROJECT SCOPE STATEMENT

Project Name

AI Learning Environment (working name)

Project Deliverables

1. Project management and course deliverables

1.1 Planning documents

Workshop 1 to 3 documents: project ideation, competitor analysis, user analysis, business model analysis, business case, charter, stakeholder analysis and plan, requirements, scope statement, schedules, roles and RACI chart.

1.2 Pitches and vlogs

Three-project pitch (15 Sep 2026), final pitch at Cultivator HQ and Vlog 1 (13 Oct 2026), bazaar day presentation (24 Nov 2026), Vlog 2 (1 Dec 2026), and the ENSE 477 final presentation.

1.3 Project tracking

GitHub repository, Kanban board, user story map, Gantt chart and weekly scrum updates following the course scrum standard.

2. Design and usability

2.1 Low-fidelity prototype

Paper sketches or wireframes of the main flow: add a course folder, open a PDF, select a passage, ask a question, read the explanation with its page reference, generate flashcards.

2.2 High-fidelity prototype

Clickable Figma prototype of the same flow, including the model picker and quiz screens.

2.3 Usability evaluation

Usability questionnaire (task list, SUS questions and open questions), sessions with at least 5 students on 20 October, analysis of results and a redesign.

2.4 System architecture

Diagrams of the desktop app, AI layer, server proxy and local database, done before coding starts on 27 October.

3. Desktop app: study core (Fall 2026)	3.1 Course folders and PDF reader Add and remove local course folders, list PDFs by course, open them in a PDF.js reader with navigation, zoom, search and passage selection (FR-1 to FR-3, FR-12). 3.2 AI layer, server proxy and model picker Model-independent AI interface with adapters for free and paid providers, a server proxy that holds API keys and rate-limits requests, and the model picker sorted by price (FR-6, FR-11, TR-4 to TR-6). 3.3 Page-anchored explanations Ask by text about a selection and receive a streamed explanation with a clickable file and page reference (FR-4, FR-5, FR-10). 3.4 Local storage, flashcards and quizzes SQLite storage for course history; flashcard generation and editing; multiple-choice quiz generation with instant marking (FR-7 to FR-9).
4. Desktop app: Pro and review (Winter 2027)	4.1 Accounts, Pro subscription and usage limits Sign-up and sign-in, Pro subscription through a payment processor, monthly usage limits with fallback to free models (FR-13 to FR-15). 4.2 Spaced review Exam dates per course and a daily flashcard review queue scheduled before each exam (FR-16). 4.3 Voice input and text recognition Ask questions by speaking, and convert scanned pages to text (FR-17, FR-18).
5. iPad version (Winter 2027)	5.1 iPad app beta Read PDFs, ask questions and review flashcards on an iPad, using the same AI layer and server (FR-19).
6. Testing, pilot and release	6.1 Accuracy and performance testing A 20-question test set and a 30-explanation accuracy review against the source pages, plus speed checks against PR-1 to PR-5.

6.2 Design-partner pilot

At least 15 students use the app for at least 3 weeks; weekly activity and willingness to pay for Pro are measured.

6.3 Final release and showcase

Installable desktop app, user guide, final presentation and capstone showcase in April 2027.

Project Exclusions

- Android, iPhone, Linux and web versions.
 - File types other than PDF, such as Word, PowerPoint or handwritten notes.
 - Uploading, syncing or storing course files on any server.
 - Sharing decks or notes between students, or other collaboration features.
 - Automatic download from UR Courses or any other learning management system.
 - Building or training the team's own AI models. A fully on-device model is a possible experiment in ENSE 477, not a committed deliverable.
 - Handwriting and annotation tools.
 - Campus or department licences and instructor dashboards (a business model option, not built in the capstone).
 - Answers drawn from web search or sources outside the student's own files.
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